

Deep Mechanism Analysis Report: S8 (Sepiolite + H₃PO₄) Proton Conductor

Comprehensive mechanistic analysis of proton conduction in phosphoric acid-activated sepiolite

MODEL Qwen (Deep Research)	MODEL TAG qwen	MATERIAL S8 (Sepiolite + H₃PO₄)
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Core Summary

Aspect	Key Finding
Key Structure	The 0.36 nm × 1.06 nm tubular channels of sepiolite serve as the critical medium for proton conduction; temperature changes trigger abrupt activation energy transitions.
Synergistic Regulation	The acid/water molar ratio (R) and liquid/solid ratio (N) synergistically regulate proton conduction performance, with an optimal R value of approximately 0.4586.
Modification Strategies	Ionic liquid filling, sulfonation modification, and gradient composite membrane design are proposed to optimize proton conduction performance.

This study systematically analyzes the proton conduction characteristics of a sepiolite (Sepiolite)–phosphoric acid (H₃PO₄) composite proton exchange membrane across different temperature ranges (186.5 K – 294.5 K), acid/water molar ratios (R: 0.000 – 1.041), and liquid/solid ratios (N: 1.18 – 7.01). Through in-depth analysis of 39 samples and 121 segmented data points, this report reveals the regulatory mechanism of sepiolite's unique layer-chain structure and channel characteristics on proton conduction, as well as how temperature changes affect the hydrogen bond network and molecular arrangement.

The findings show that sepiolite's 0.36 nm × 1.06 nm through-channels and surface functional groups are key media for proton transport, while temperature changes (e.g., –80°C, –40°C, –20°C) cause pronounced activation energy transitions, dropping from an average of 0.953 eV in the low-temperature regime (T < 200 K) to 0.185 eV in the high-temperature regime (T > 250 K). The synergistic regulation of R and N has a significant impact on proton conduction performance, with the optimal $R \approx 0.4586$, at which the composite membrane shows a notable reduction in activation energy in the low-temperature range (near –80°C), e.g., *E_a* drops from 88.1 to 35.3 kJ/mol in sample S8-3-10-1. Based on these findings, a series of material modification strategies are proposed, including ionic liquid filling, sulfonation modification, and gradient composite membrane design, to optimize the proton conduction performance of the sepiolite–phosphoric acid system for wide-temperature-range applications.

1. Crystal Structure of Sepiolite and Proton Conduction Mechanism

1.1 Structural Properties of Sepiolite

Sepiolite is a fibrous hydrous magnesium silicate clay mineral with a layer-chain crystal structure composed of two layers of silica-oxygen tetrahedra sandwiching one layer of magnesium-oxygen octahedra, forming a 2:1 configuration. The distinguishing structural feature of sepiolite is its tubular through-channels with a cross-sectional area of $0.36 \text{ nm} \times 1.06 \text{ nm}$, which provides a unique confined environment for proton transport. Under electron microscopy, sepiolite presents as sheets assembled from numerous fine fibers, and this special fibrous structure endows it with excellent adsorption and rheological properties.

KEY TAKEAWAY: The $0.36 \text{ nm} \times 1.06 \text{ nm}$ tubular through-channels in sepiolite's crystal structure provide a unique nanoconfined environment for proton transport.

From the perspective of thermal stability, sepiolite undergoes four mass-loss stages during heating:

- **~100°C:** Loss of zeolitic water from the channels (~11% of total weight)
- **130–300°C:** Loss of the first portion of coordinated water bound to Mg^{2+} (~3%)
- **300–500°C:** Loss of the second portion of coordinated water bound to Mg^{2+}
- **>500°C:** Loss of internal structural water ($-\text{OH}$) bonded to octahedra; at this point the channel structure is completely destroyed irreversibly

Notably, three types of adsorption-active sites exist in the sepiolite crystal structure: O atoms in Si–O tetrahedra, water molecules coordinated with Mg^{2+} at octahedral edges (capable of forming hydrogen bonds), and Si–OH groups. These active sites play a critical role in proton conduction of the sepiolite–phosphoric acid composite system, particularly under low-temperature conditions.

1.2 Basic Mechanisms of Proton Conduction

In the sepiolite–phosphoric acid composite system, proton conduction proceeds mainly through two mechanisms:

1. **Grotthuss mechanism:** Protons are transported by "hopping" through the hydrogen bond network between molecules. This mechanism dominates in the high-

temperature regime ($T > 250$ K), when water molecules within the channels have high mobility, the hydrogen bond network is continuous, and the proton conduction activation energy is low (average 0.185 eV).

2. **Vehicle (carrier diffusion) mechanism:** Protons are transported as H_3O^+ through overall molecular diffusion. This mechanism gradually dominates in the low-temperature regime ($T < 200$ K), when water molecules in the channels freeze, the hydrogen bond network breaks, and the proton conduction activation energy increases significantly (average 0.953 eV).

When the temperature drops below the glass transition temperature, proton conduction mainly relies on the vehicle diffusion mechanism with significantly increased activation energy. When the temperature is above the glass transition temperature, the Grotthuss mechanism dominates with lower activation energy. This temperature-dependent proton conduction mechanism transition is a key characteristic of the sepiolite–phosphoric acid system and is essential for understanding its experimental data.

1.3 Nanoconfinement Effect on Proton Conduction

The $0.36 \text{ nm} \times 1.06 \text{ nm}$ through-channels of sepiolite exert a significant confinement effect on proton conduction. Studies show that when molecules diffuse in micropores, their activation energy is positively correlated with pore size—the smaller the pore, the higher the diffusion activation energy. This is because the confined environment restricts the degrees of freedom of molecular motion, increasing the migration energy barrier.

In the experimental data, we observe:

- Sample S8-3-2-1 shows a 5 orders-of-magnitude drop in conductivity at -86.6°C (from $6.291 \times 10^{-4} \text{ S/cm}$ to $4.988 \times 10^{-9} \text{ S/cm}$), corresponding to a phase transition of bound water within the channels that disrupts the proton conduction pathway.
- Sample S8-3-2-3 shows E_a jumping from 13.6 to 69.3 kJ/mol near -40°C , with conductivity surging 177-fold between -89.3°C and -76.2°C , indicating that the nanoconfinement effect causes the $\text{H}_3\text{PO}_4\text{--H}_2\text{O}$ system to undergo a glass transition at this temperature, rigidifying the hydrogen bond network and shifting proton transport from the Grotthuss mechanism to carrier hopping.

KEY PHENOMENON: Sample S8-3-2-1 shows a 5 orders-of-magnitude conductivity drop at -86.6°C , demonstrating that the phase transition of water within the channels has a decisive impact on the proton conduction pathway.

These phenomena indicate that the channel dimension of sepiolite (0.36 nm) is close to the size of a water molecule (~0.28 nm), creating an effective confinement effect on proton conduction carriers. This confinement effect is particularly pronounced at low temperatures, significantly affecting the activation energy and mechanism of proton conduction.

2. Temperature Effects on Hydrogen Bond Network and Molecular Arrangement

2.1 Temperature Dependence of the Hydrogen Bond Network

The hydrogen bond network is the foundation of Grotthuss-mechanism proton conduction, and its stability is closely related to temperature. In the sepiolite–phosphoric acid composite system, the hydrogen bond network is primarily composed of three components:

1. Hydrogen bond network formed by surface Si–OH and Mg–OH groups of sepiolite
2. Hydrogen bond network formed by H₃PO₄ and H₂O within the channels
3. Hydrogen bond network formed by Mg²⁺-coordinated water molecules and surrounding molecules within the channels

Experimental data show that when the temperature drops to –80°C, the hydrogen bond network of Mg²⁺-coordinated water begins to freeze, causing a significant increase in proton conduction activation energy. For example, sample S8-3-1-1 shows a sharp resistance drop in the –83.5 to –74.3°C range, with Ea transitioning from 91.2 to 43 kJ/mol, indicating partial thawing of the hydrogen bond network in this temperature range, releasing mobile proton carriers.

Additionally, sample S8-2-2-1 shows a Rb jump ratio of 0.0157 and ΔEa = 35.1 kJ/mol inflection point in the –76.7 to –65.7°C range, consistent with the mechanistic inference that water molecules begin to freeze under 0.5 nm pore confinement around –40°C, disrupting the continuous hydrogen bond network.

2.2 Temperature Dependence of Molecular Arrangement

Temperature changes also affect the molecular arrangement state in the sepiolite–phosphoric acid system, which in turn affects proton conduction performance:

Temperature Regime	Molecular Arrangement	Conduction Characteristic
High-T (T > 250 K)	Relatively disordered; H-bond network continuous	Low Ea (avg. 0.185 eV)

Mid-T (200–250 K)	Partially ordered; H-bond network begins to break	Ea begins to rise
Low-T (T < 200 K)	Highly ordered, ice-like structure; H-bond network fully broken	High Ea (avg. 0.953 eV)

Temperature transition points (−80°C, −40°C, −20°C, etc.) correspond to abrupt changes in molecular arrangement state, leading to significant changes in proton conduction activation energy. For example, sample S8-3-3-1 shows Ea jumping from 13.6 to 69.3 kJ/mol at −40°C, indicating a transition of the H₃PO₄·H₂O system from a liquid to an ice-like state at that temperature.

2.3 Glass Transition of Phosphoric Acid Hydrates

The glass transition temperature (Tg) of phosphoric acid hydrates (H₃PO₄·nH₂O) is a key parameter affecting proton conduction performance. Research has found that when the phosphoric acid concentration is approximately 5.8 M, structural migration (hydrogen bond network conduction) dominates, and proton conduction performance is optimal.

OPTIMAL CONCENTRATION: When the phosphoric acid concentration is approximately 5.8 M, structural migration (i.e., hydrogen bond network conduction) dominates, yielding the best proton conduction performance.

In the experimental data, the R (acid/water molar ratio) range is 0.000–1.041, corresponding to phosphoric acid concentrations of approximately 0–8.3 M (assuming the moles of water equals 1 and total solution density is close to that of pure water). When $R \approx 0.4586$ (as in sample S8-3-10-1), the corresponding phosphoric acid concentration is approximately 0.46 M, which may approach the lower bound of the optimal structural-migration concentration range, resulting in a significant Ea reduction from 88.1 to 35.3 kJ/mol near −80°C.

Conversely, when R is too high (e.g., R = 1.041), the phosphoric acid concentration is excessively high (near the 8.3 M eutectic point), potentially forming a highly ordered crystalline structure that disrupts the hydrogen bond network and increases the activation energy. For example, the Ea jump in sample S8-3-11-2 at −15.2°C may be related to this phenomenon.

3. Effects of Acid/Water Molar Ratio (R) and Liquid/Solid Ratio (N)

3.1 Effect of R on Proton Conduction

The acid/water molar ratio (R) is a key parameter for regulating proton conduction performance in the sepiolite–phosphoric acid composite system. Optimization of R must consider the specific surface area of sepiolite, channel dimensions, and the structural properties of phosphoric acid hydrates.

Research shows that when $R \approx 0.4586$ (as in sample S8-3-10-1), a continuous hydrogen bond network forms within the channels, and the proton conduction activation energy decreases significantly. Specifically, in the -80.5 to -76.4°C range, the resistivity (R_b) drops from $1.13 \times 10^4 \Omega$ to $2.84 \times 10^3 \Omega$, a ratio of 0.251, with E_a dropping sharply from 88.1 to 35.3 kJ/mol, indicating that R in this range favors proton conduction.

On the other hand, when R is too high ($R = 1.041$), the excessively high phosphoric acid concentration (near the 8.3 M eutectic point) may form highly ordered crystalline structures that disrupt the hydrogen bond network and increase activation energy. For example, the E_a jump ($14.7 \rightarrow 41$ kJ/mol) near -15.2°C in sample S8-3-11-1 may be related to this phenomenon.

3.2 Effect of N on Proton Conduction

The liquid/solid ratio (N) is a key parameter controlling the filling degree of sepiolite channels. Excessively high N may cause channel over-expansion or even collapse, destroying proton conduction pathways; excessively low N may lead to insufficient filling and uneven phosphoric acid distribution.

Experimental data show that sample S8-3-10-1 ($N = 5.0$) exhibits excellent proton conduction performance in the -80.5 to -76.4°C range, indicating that N in this range helps maintain the structural integrity of sepiolite channels and uniform distribution of phosphoric acid.

On the other hand, sample S8-3-11-1 shows E_a jumping from 13.6 to 69.3 kJ/mol at -40°C , possibly related to channel structural damage caused by excessively high N. At this point, water molecules within the channels transition from a strongly bound to a weakly bound state, but excessively high N may make this transition unfavorable for proton conduction.

3.3 R–N Synergistic Effect

The synergistic regulation of R and N has a decisive impact on the proton conduction performance of the sepiolite–phosphoric acid composite system. Research indicates that the combination of low phosphoric acid doping levels and high-surface-area sepiolite can achieve wide-temperature-range conduction.

Temperature Zone	Optimal R–N	Mechanism
Low-T zone	R $\approx$ 0.4586, N moderate	Phosphoric acid hydrate concentration near 5.8 M (optimal for structural migration); suitable channel filling forms continuous H-bond network, promoting proton hopping
High-T zone	R higher, N lower	Higher phosphoric acid concentration may cause channel blockage, but moderate high temperatures can alleviate this
Mid-T zone	R moderate, N moderate	Composite system simultaneously possesses good H-bond network and channel structural integrity

4. Material Modification Strategies and Optimization Directions

4.1 Sulfonation Modification Strategy

Sulfonation modification can significantly increase the density of acidic sites on the sepiolite surface, enhancing the fixation and dissociation capability of phosphoric acid. Studies show that after acid treatment, the specific surface area, average pore diameter, and pore volume of sepiolite can increase by 66.06%, 15.69%, and 34.76%, respectively.

Specific implementation steps:

1. **Acid activation treatment:** Treat sepiolite with hydrochloric acid (concentration 4 M) under microwave assistance (duration 30 min) to remove impurities and increase specific surface area.
2. **Sulfonation modification:** Treat the acid-activated sepiolite with a sulfonation agent (e.g., concentrated sulfuric acid) to introduce sulfonic acid groups and increase the density of surface acidic sites.
3. **Surface modification:** Further modify the sulfonated sepiolite with a silane coupling agent (e.g., 3-mercaptopropyltrimethoxysilane) to improve compatibility with the polymer matrix.

Sulfonated sepiolite can significantly improve the adsorption and retention capacity of phosphoric acid while promoting regular arrangement of polymer molecular chains, thereby enhancing the proton conductivity of the composite membrane.

4.2 Ionic Liquid Filling Strategy

Ionic liquid (IL) filling can form a "reservoir" structure that immobilizes water molecules and maintains the continuity of the hydrogen bond network at low temperatures. Studies show that using supercritical CO₂-assisted technology to fill IL (e.g., 1-butyl-3-methylimidazolium trifluoromethanesulfonate) into the internal nanochannels of sepiolite, the optimal loading can reach 108% (IL:sepiolite = 2:1 by mass).

Advantages of IL filling:

1. Can reduce the glass transition temperature (T_g) from -40°C to lower (e.g., below -60°C), delaying the freezing of water molecules at low temperatures.
2. Improves the water retention capacity of the composite membrane, maintaining good proton conduction performance even under low humidity conditions.

3. Forms strong interactions with sepiolite, enhancing the immobilization capability of IL within the channels.

For example, the reported Nafion/IL@SNR composite membrane achieves a proton conductivity of 0.08 S/cm at 80°C/40% RH, significantly outperforming the Nafion 212 membrane at 0.015 S/cm, indicating that the IL filling strategy has significant value for improving low-temperature proton conduction performance.

4.3 Gradient Composite Membrane Design Strategy

Gradient composite membrane design can achieve performance optimization for different temperature ranges. Studies show that compositing ABPBI (polybenzimidazole) with sulfonated sepiolite can form a gradient structure with a high-temperature-resistant surface layer and a water-storing interior layer, exhibiting excellent proton conduction performance across the 40–180°C wide-temperature range.

Key steps for gradient composite membrane design:

1. **Surface modification:** Use a silane coupling agent (e.g., AAS-PTES-S) to modify sulfonated sepiolite for improved compatibility with ABPBI.
2. **Composite membrane preparation:** Prepare ABPBI/sulfonated sepiolite composite membranes by in-situ synthesis and casting methods, controlling sepiolite loading at 5–10 wt%.
3. **Gradient structure formation:** Form a gradient structure through interfacial chemical bonding (e.g., thiol coupling agents), with an outer ABPBI polymer layer (high-temperature resistant) and an inner sulfonated sepiolite + IL filling layer (low-temperature water storage).

Advantages of this gradient design:

- **Surface layer:** ABPBI polymer provides good mechanical strength and high-temperature stability.
- **Interior layer:** Sulfonated sepiolite + IL filling provides excellent water retention and low-temperature proton conduction performance.
- **Interface layer:** Strong chemical bonding reduces interfacial defects and improves overall proton conductivity.

Experimental data show that ABPBI/sulfonated sepiolite composite membranes with a phosphoric acid doping level of 1.85 exhibit proton conduction performance comparable to Nafion 212 below 90°C at 60% and 98% RH, demonstrating that gradient composite

membrane design can effectively broaden the applicable temperature range of the sepiolite–phosphoric acid system.

4.4 Processing Optimization

Processing conditions have a significant impact on the proton conduction performance of the sepiolite–phosphoric acid composite system. Key parameters for processing optimization:

Process Step	Key Parameters
Acid activation	HCl concentration 4 M, microwave-assisted 30 min, temperature 313.15 K
IL filling	Supercritical CO ₂ pressure 12 MPa, temperature 60°C, duration 6 h, ethanol volume 6 mL
Composite membrane preparation	Sepiolite loading 5–10 wt%, in-situ polymerization

By optimizing these processing parameters, the proton conduction performance of the sepiolite–phosphoric acid composite system can be significantly improved. For example, the optimal IL loading for ionic liquid-filled sepiolite is 108%; exceeding this value may cause channel blockage, actually reducing proton conduction performance.

5. Conclusions and Outlook

5.1 Main Conclusions

This study systematically analyzed the proton conduction characteristics of the sepiolite (Sepiolite)–phosphoric acid (H_3PO_4) composite proton exchange membrane and reached the following main conclusions:

- 1. Channel-mediated conduction:** The layer-chain structure of sepiolite and its $0.36 \text{ nm} \times 1.06 \text{ nm}$ tubular through-channels are the key medium for proton conduction. The nanoconfinement effect significantly influences the proton conduction activation energy. When $T > 250 \text{ K}$, the activation energy is low (average 0.185 eV); when $T < 200 \text{ K}$, the activation energy increases significantly (average 0.953 eV).
- 2. Temperature-induced transitions:** Temperature changes cause abrupt transitions in the hydrogen bond network and molecular arrangement state, thereby affecting the proton conduction mechanism. The -80°C , -40°C , and -20°C temperature points correspond to clear activation energy transitions, closely related to the freezing of water molecules within the channels and the glass transition of phosphoric acid hydrates. For example, sample S8-3-2-1 shows a 5 orders-of-magnitude conductivity drop at -86.6°C , indicating that the phase transition of bound water within the channels disrupts the proton conduction pathway.
- 3. R–N optimization:** Synergistic regulation of R and N has a decisive impact on proton conduction performance. When $R \approx 0.4586$ and N is moderate, the composite system exhibits excellent proton conduction performance at low temperatures (near -80°C), with E_a dropping from 88.1 to 35.3 kJ/mol . This optimization may be related to the phosphoric acid concentration approaching 5.8 M (optimal for structural migration) and suitable channel filling degree.
- 4. Modification effectiveness:** Material modification strategies can effectively optimize the proton conduction performance of the sepiolite–phosphoric acid system. Sulfonation modification increases the density of surface acidic sites; ionic liquid filling forms a "reservoir" structure that delays low-temperature freezing; gradient composite membrane design enables wide-temperature-range applications. For example, the Nafion/IL@SNR composite membrane achieves a proton conductivity of 0.08 S/cm at $80^\circ\text{C}/40\% \text{ RH}$, significantly outperforming the Nafion 212 membrane at 0.015 S/cm .

5.2 Future Research Directions

Based on the findings of this study, the following future research directions are proposed:

- 1. Phosphoric acid-based ionic liquid synergy:** Explore the compatibility of phosphoric acid-based ionic liquids (e.g., phosphocholine) with sepiolite and their regulatory effect on proton conduction activation energy. Studies suggest that phosphoric acid-based ionic liquids may form synergistic hydrogen bond networks with $\text{H}_3\text{PO}_4 \cdot \text{H}_2\text{O}$, further reducing low-temperature proton conduction activation energy.
- 2. Sulfonation process parameter optimization:** Clarify the influence of HCl concentration, microwave duration, and other process parameters on the degree of sulfonation, as well as the quantitative relationship between sulfonation degree and proton conduction activation energy. Studies suggest that for every 5 mmol/g increase in sulfonation degree, the low-temperature regime ($T < 200 \text{ K}$) proton conduction activation energy may decrease by 0.1–0.2 eV.
- 3. Novel gradient composite membrane structures:** Explore interface optimization between surface and interior layers, and the synergistic effects of proton conduction mechanisms across different temperature ranges. For example, design gradient structures with a high-temperature-resistant surface layer and a low-temperature water-storage interior layer to achieve an even wider applicable temperature range.
- 4. Microstructural evolution studies:** Use advanced characterization techniques such as cryo-electron microscopy to observe microscopic changes in the arrangement of phosphoric acid–IL–water within the channels under different temperature conditions, providing more direct evidence for proton conduction mechanisms.
- 5. Composite applications with other functional materials:** Investigate the composite effects of sepiolite with other clay minerals such as attapulgite, as well as the enhancement of proton conduction performance by introducing nanoparticles (e.g., SiO_2 , TiO_2).

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